

Table 1: Simulated data: MSE and LINEX ($a=1$) across key horizons. Panel A: mean (std) over 100 seeds; Panel B: mean (SE) over 100 seeds (seeds with $|\text{LINEX}| \geq 10^4$ excluded from Panel B). 500 Monte Carlo paths, $T=600$, burn-in 100. HHI and Δ_{best} (percentage improvement over the ex-post best individual forecaster) are reported at $h=8$.

Method	Loss (mean / std)					Avg. HHI		$\Delta_{\text{best},8}$
	$h=1$	$h=4$	$h=8$	$h=12$	$h=16$	$h=4$	$h=8$	
<i>Panel A : MSE — mean (std)</i>								
MWUM _{Conc}	0.932 (0.12)	2.273 (0.38)	2.953 (0.63)	3.211 (0.74)	3.342 (0.81)	0.274	0.314	+28.5%
MWUM _{Both}	0.917 (0.12)	2.388 (0.39)	2.985 (0.62)	3.206 (0.72)	3.329 (0.89)	0.174	0.220	+27.6%
MWUM _{Van} [†]	0.917 (0.12)	3.210 (0.55)	4.196 (0.92)	4.724 (2.79)	4.389 (1.12)	0.294	0.985 [†]	−1.3%
OGD _{Van}	0.923 (0.12)	2.499 (0.41)	3.324 (0.68)	3.595 (0.85)	3.722 (0.95)	0.222	0.189	+19.3%
OGD _{Both}	0.923 (0.12)	2.476 (0.41)	3.313 (0.67)	3.598 (0.82)	3.720 (0.94)	0.144	0.184	+19.6%
OGD _{Conc}	0.935 (0.12)	2.554 (0.43)	3.499 (0.79)	3.910 (0.97)	4.136 (1.08)	0.087	0.098	+15.5%
RL- κ	1.036 (0.15)	2.381 (0.39)	3.096 (0.66)	3.359 (0.75)	3.487 (0.80)	0.372	0.383	+24.9%
RL-Rule	0.928 (0.12)	2.872 (0.55)	3.731 (0.90)	4.130 (1.05)	4.298 (1.16)	0.128	0.206	+9.9%
Mean	0.937 (0.12)	3.285 (0.56)	4.736 (1.05)	5.337 (1.34)	5.588 (1.50)	0.056	0.056	−14.4%
Median	0.924 (0.12)	3.262 (0.57)	4.525 (1.03)	4.967 (1.28)	5.082 (1.41)	—	—	−9.1%
<i>Panel B : LINEX ($a=1$) — mean (SE) over 100 seeds</i>								
MWUM _{Conc}	1.530 (0.35)	6.787 (1.33)	17.08 (4.07)	30.40 (9.88)	52.14 (18.38)	0.274	0.314	+34.0%
MWUM _{Both}	1.234 (0.19)	6.821 (1.33)	16.75 (4.00)	38.99 (11.30)	53.44 (18.34)	0.174	0.220	+33.9%
MWUM _{Van} [†]	1.246 (0.19)	12.75 (2.94)	33.35 (7.66)	34.52 (6.94)	70.40 (22.89)	0.294	0.985	−35.1%
OGD _{Van}	1.495 (0.34)	7.227 (1.49)	16.73 (3.95)	99.98 (72.67)	55.82 (16.85)	0.222	0.189	+32.2%
OGD _{Both}	1.495 (0.34)	6.856 (1.35)	16.65 (3.96)	37.22 (11.21)	56.13 (17.00)	0.144	0.184	+33.1%
OGD _{Conc}	1.508 (0.35)	7.122 (1.34)	17.43 (4.18)	24.53 (4.63)	59.49 (18.99)	0.087	0.098	+31.5%
RL- κ	1.848 (0.38)	8.147 (1.60)	18.83 (4.16)	32.85 (10.00)	57.60 (18.32)	0.372	0.383	+23.1%
RL-Rule	1.187 (0.13)	22.80 (6.84)	96.78 (60.42)	103.26 (34.73)	272.70 (90.74)	0.128	0.206	−91.6%
Mean	1.145 (0.13)	24.37 (9.54)	134.61 (89.99)	47.25 (6.45)	77.04 (19.06)	0.056	0.056	−746.6%
Median	1.224 (0.15)	26.07 (10.64)	73.58 (31.05)	41.75 (5.49)	53.87 (9.32)	—	—	−272.1%

Notes: Equal-weight HHI baseline: $1/18 \approx 0.056$. 100 seeds span a spectrum of macro volatility regimes. Panel B SE = $\text{std}/\sqrt{n}$ after excluding seeds with $|\text{LINEX}| \geq 10^4$; blowup counts by method available in the replication notebook. [†] MWUM_{Van} weight collapse: HHI $\rightarrow 1$ at $h \geq 8$; both entries excluded from comparisons.

Table 2: Survey of Professional Forecasters (SPF): cumulative MSE and LINEX ($a=1$) for two- and three-quarter-ahead CPI inflation forecasts ($T=34$ quarters, 12 panellists). Δ_{best} : percentage improvement in MSE over the ex-post best individual panellist (positive = outperforms oracle). [lo, hi]: 95 % moving-block-bootstrap CI ($B=4,000$, $l=4$ quarters).

Method	CPI2					CPI3				
	MSE	LINEX	HHI	Δ	[lo, hi]	MSE	LINEX	HHI	Δ	[lo, hi]
MWUM _{Conc}	2.242	0.499	0.172	-41.4%	[-107, +2]	4.429	0.946	0.294	+9.0% [†]	[-5.3, +23.6]
MWUM _{Both}	2.439	0.536	0.083	-53.9%	[-136, -1]	6.700	1.733	0.083	-37.7%	[-102, +6]
MWUM _{Van}	2.439	0.536	0.083	-53.9%	[-136, -1]	6.698	1.732	0.083	-37.7%	[-102, +6]
OGD _{Van}	2.353	0.532	0.087	-48.5%	[-128, +3]	6.172	1.525	0.086	-26.8%	[-75, +5]
OGD _{Both}	2.352	0.532	0.087	-48.4%	[-128, +3]	6.170	1.524	0.086	-26.8%	[-75, +5]
OGD _{Conc}	2.416	0.535	0.084	-52.4%	[-135, 0]	6.313	1.649	0.084	-29.8%	[-82, +5]
RL- κ	2.659	0.542	0.306	-67.8%	[-135, -19]	5.287	0.840	0.726	-8.7%	[-56, +22]
RL-Rule	2.368	0.538	0.087	-49.4%	[-124, -1]	8.507	1.897	0.098	-74.8%	[-216, +6]
Mean	2.452	0.537	0.083	-54.7%	[-139, -1]	6.848	1.886	0.083	-40.7%	[-109, +6]
Median	2.338	0.511	—	-47.5%	[-127, -1]	6.537	1.703	—	-34.3%	[-95, +8]

Notes: Single time series per horizon ($n=1$, $T=34$ quarters, 2014Q2–2022Q3). CI columns: 95 % percentile bootstrap, moving blocks (Künsch 1989), $B=4,000$ resamples, $l=\lceil T^{1/3} \rceil=4$ quarters; wide CIs reflect the limited power available from a single short series. [†] MWUM_{Conc} at CPI3 is the only positive Δ ; Diebold–Mariano test (Newey–West SE, $L=4$ lags) gives one-sided $p=0.08$. Equal-weight HHI baseline: $1/12 \approx 0.083$. CPI1 omitted: MSE ≈ 0.203 , LINEX ≈ 0.025 (indistinguishable across methods).

Table 3: M3 monthly macroeconomic competition (301 series, $T=18$): head-to-head comparison of representative methods. LINEX not reported (exponential overflow at M3 scale).

	MWUM _{Conc}	RL- κ	RL-Rule	Median	Mean	OGD [†]
MSE ($\times 10^3$)	478.03	478.01	624.44	683.25	696.63	1,039.0
MAE	270.4	270.4	353.5	428.2	421.9	430.5
Impr. vs Mean	+31.4%	+31.4%	+10.4%	+1.9%	0%	-49.2%

Notes: MWUM_{Both} and MWUM_{Van} yield MSE = 478.43×10^3 ($< 0.1\%$ from MWUM_{Conc}) and are omitted. All three OGD variants yield identical results to two decimal places; one shown. RL- κ and MWUM_{Conc} differ by just 18 MSE units (0.004%) — both converge to near-identical near-degenerate concentration (HHI ≈ 0.89) under $T=18$, but MWUM’s scale-invariant updates select the correct forecaster while OGD step sizes — calibrated at SPF scale — diverge. Impr. vs Mean: $(696.63 - \text{method})/696.63$; positive = better than equal-weight mean.